DAY 22 | HANDS-ON ACTIVITIES

Context Engineering and RAG: Problem Statements

Seven problems using the sample employee handbook in the notebook. All run offline; add a key for real model answers.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Map the Context

Level: Warm-up | **Concepts:** context components

For an HR assistant chatbot, list everything that should go into the context for the question "Can I carry forward my leave?" and give each part a priority and a token budget out of 4,000.

Sample run / expected result

```
rules 300 (1), question 50 (1), leave chunks 1500 (2), user profile 100 (2), history summary
```

Hints

- Always reserve space for the answer.

Problem 2: Three Chunkers

Level: Easy | **Concepts:** fixed, paragraph, recursive

Chunk the handbook with fixed-size (300 chars, 50 overlap), paragraph and header-based chunkers. Print the number of chunks and the average length for each.

Sample run / expected result

```
fixed: 9 chunks, avg 290
paragraph: 12 chunks, avg 160
headers: 5 chunks, avg 350
```

Hints

- Header-based: split on lines starting with ##.

Problem 3: Spot the Broken Chunk

Level: Easy | **Concepts:** chunk boundaries

Find a fixed-size chunk that cuts a sentence in half. Explain what question would fail because of it, and fix it with overlap or sentence boundaries.

Sample run / expected result

```
Chunk 4 ends: "...expires on 31 March of the follo"
```

Hints

- Search for chunks that do not end with a full stop.

Problem 4: Context Budget Assembler

Level: Medium | **Concepts:** token budgets, priorities

Write assemble(parts, budget) that adds parts by priority, skips or truncates what does not fit, and reports what was dropped.

Sample run / expected result

```
Used 3,940 / 4,000 tokens. Dropped: old history (420 tokens)
```

Hints

- Estimate tokens as len(text) / 4 if tiktoken is not available.

Problem 5: Grounded Answers

Level: Medium | **Concepts:** RAG prompt, citations

Retrieve the top 3 chunks for five questions and answer with a grounded prompt that cites chunk ids. Include one question the handbook cannot answer.

Sample run / expected result

```
Q: How many sick days? -> 8 days [c6]
Q: Is there a gym? -> I don't know
```

Hints

- The "I don't know" case is as important as the correct ones.

Problem 6: Contextual Headers

Level: Challenging | **Concepts:** chunk metadata

Prefix every chunk with its section title and compare retrieval for "When does carried forward leave expire?" with and without headers.

Sample run / expected result

```
Without headers: correct chunk ranked 2
With headers: ranked 1
```

Hints

- Keep the header in the text you embed.

Problem 7: Conversation Memory

Level: Stretch | **Concepts:** history compression

Simulate a 12-turn conversation. Keep the last 3 turns verbatim and replace older turns with a summary (offline: first sentence of each). Show the token saving.

Sample run / expected result

```
Full history: 1,850 tokens
Compressed: 620 tokens
```

Hints

- Summaries lose detail; keep facts the user stated explicitly.

Ganbatte (頑張って): do your best